

A one-edge flag-complex counterexample to conjecture 00000002712

earthking11

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Scope and standard definitions

The conjecture proposes that the g -vectors of flag simplicial complexes are nonnegative. It does not require the complexes to be homology spheres, boundaries of polytopes, or closed manifolds. We use a connected, pure flag complex within that stated class: the full 1-simplex. No assertion concerning the usual sphere-restricted g -conjecture is being refuted.

For a $(d-1)$ -dimensional complex, let f_{j-1} count faces with j vertices, including $f_{-1} = 1$ for the empty face. The standard transform and truncated g -vector are

$$\sum_{i=0}^d h_i t^{d-i} = \sum_{j=0}^d f_{j-1} (t-1)^{d-j}, \quad g_0 = h_0, \quad g_i = h_i - h_{i-1} \quad (1 \leq i \leq \lfloor d/2 \rfloor).$$

Theorem 1. *The clique complex of the actual graph K_2 is connected, pure, and flag, but its standard truncated g -vector has $g_1 = -1$.*

Proof. Let the vertices be $\{0, 1\}$ with the single edge $\{0, 1\}$. Its clique complex consists exactly of the empty face, the two singleton faces, and the full edge. The face collection is downward closed; every set of pairwise adjacent vertices is a face, so it is flag. The graph is connected. The full edge is the unique facet, so the complex is pure of dimension 1, with rank $d = 2$.

Counting these actual faces gives $(f_{-1}, f_0, f_1) = (1, 2, 1)$. Therefore

$$\sum_{i=0}^2 h_i t^{2-i} = (t-1)^2 + 2(t-1) + 1 = t^2,$$

and $h = (1, 0, 0)$. The truncation range includes index 1, since $\lfloor d/2 \rfloor = 1$. Thus $g_1 = h_1 - h_0 = -1 < 0$, contradicting the claimed nonnegativity for flag complexes. $\square$

The counterexample is a simplex (a ball), not a sphere. Adding a sphere hypothesis would change the original assertion and exclude this example; the submission does not claim to solve that modified question. Refuting the stated nonnegativity already suffices, so its further suggestion of a free-resolution proof is unnecessary.

Formal verification

Lean uses Mathlib's actual `SimpleGraph` (`Fin 2`) and defines its clique faces as a finite set of vertex sets. It proves downward closure, flagness, connectedness, and facet purity. The rank is defined as the maximum actual face size and proved to equal 2; the dimension then equals 1. All entries of the face counts, including the empty-face count, are computed from this actual face collection, not assumed as an external vector.

The h -polynomial is an integer polynomial obtained by the displayed standard transform, and its coefficients define h_i . The g -vector has the exact formal index type $0, \dots, \lfloor d/2 \rfloor$. Lean proves the negative entry and a single combined counterexample theorem containing the connected, flag, pure, dimension, and negativity properties. The project compiles on Lean 4.33.1 with pinned Mathlib, without unfinished proofs, native evaluation, additional axioms, or warnings. The reported dependencies are propositional extensionality, classical choice, and quotient soundness. Computation enumerates only the four faces of this two-vertex graph; no long brute force is used.